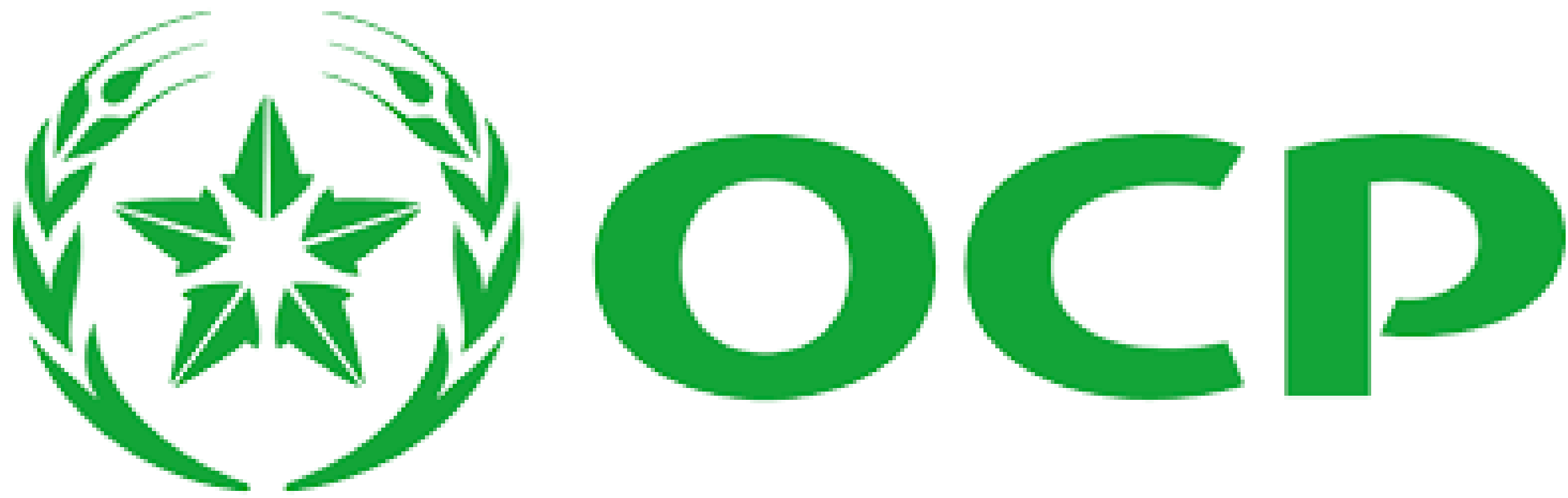




OT Cybersecurity Hardening Internship

Roadmap

Enhancing Cybersecurity in OT Environments through System and Network Hardening

Internship Subject :

Context

OS and network equipment misconfigurations expose infrastructures to vulnerabilities. This internship focuses on studying and validating hardening procedures for Windows/Linux systems and network devices (pfSense firewalls, Cisco and Moxa switches).

Objectives

- Deploy an emulated network architecture using GNS3
- Define hardening procedures based on CIS Benchmarks for Moxa/Cisco switches and firewalls
- Apply and test configurations in the emulated environment
- Validate hardening procedures for Windows Server and Linux Server
- Implement an agentless Network Access Control (NAC) policy
- Simulate real-world cyberattacks and demonstrate the effectiveness of deployed security mechanisms

Deliverables

- Emulated network architecture documentation
- Hardening procedures for network equipment and OS
- Agentless NAC policy and rules
- Attack simulation reports with result analysis
- Linux patch management strategy with rollback scenarios
- Windows/Linux server migration plans
- Final internship report and defense presentation

Internship conducted at:

OCP Group

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Project Context

Operational Technology (OT) environments rely on interconnected systems, servers, and network equipment to support critical industrial operations. Misconfigurations, **weak access controls**, and insecure services can expose these infrastructures to **cyber threats** such as **unauthorized access**, **lateral movement**, and **service disruption**.

This internship focuses on strengthening the **security posture** of an emulated OT environment through **system and network hardening** techniques. The project combines **infrastructure virtualization**, **defensive security mechanisms**, **attack simulation**, and security validation in order to reduce **the attack surface** and improve **resilience** against realistic cyber threats.

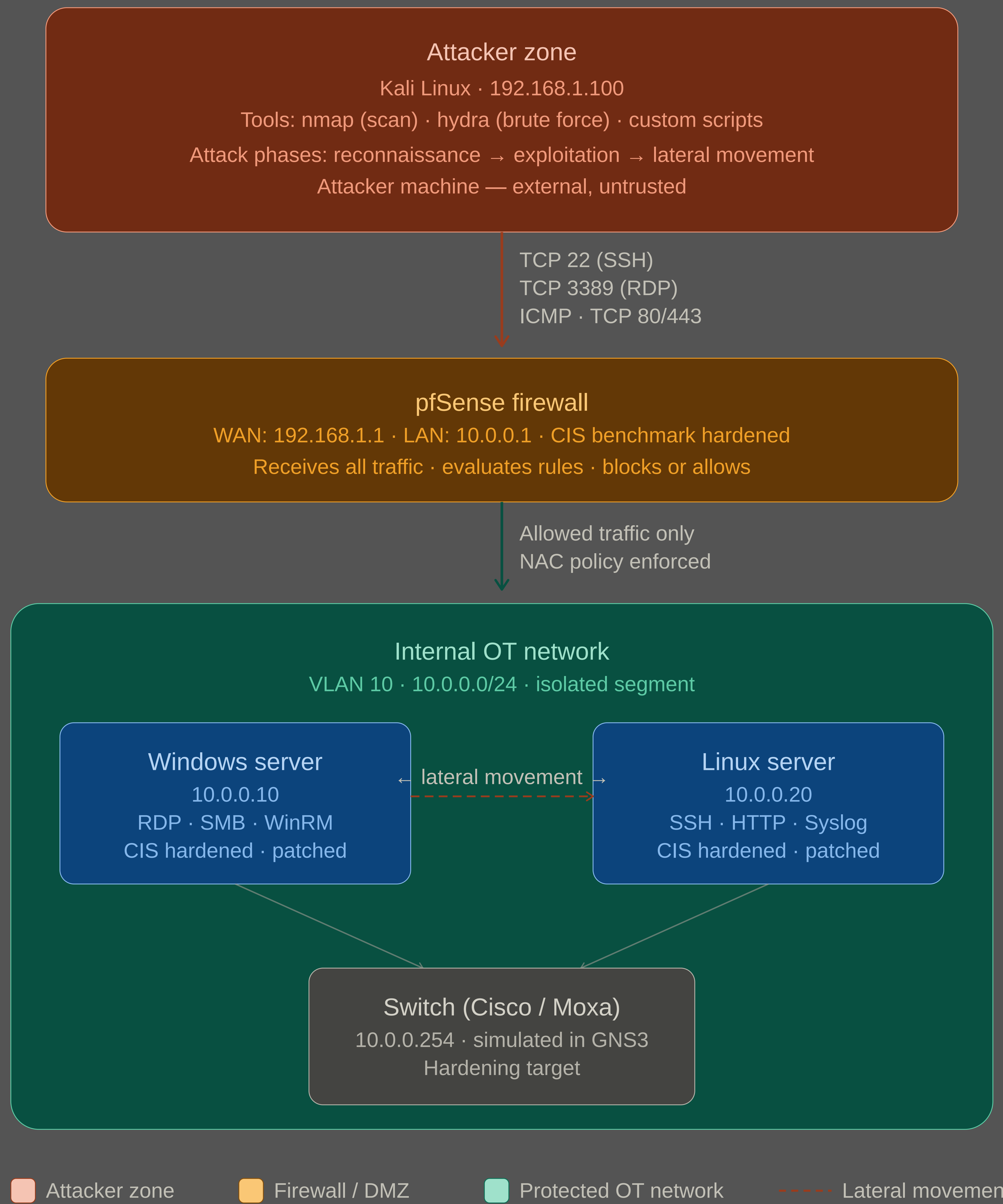
Project Objectives

The main objective of this internship is to design, secure, and validate an emulated Operational Technology (OT) environment through the implementation of defensive cybersecurity mechanisms and system hardening practices.

The project aims to:

- Design and deploy a virtual OT infrastructure using GNS3 and virtualization technologies
- Apply hardening procedures to Windows Server, Linux Server, pfSense firewalls, and network devices based on CIS Benchmarks and security best practices
- Reduce the attack surface of the infrastructure through secure configurations, segmentation, and access control mechanisms
- Implement an agentless Network Access Control (NAC) approach to restrict unauthorized access to critical network segments
- Simulate realistic cyberattack scenarios in order to evaluate the resilience of the deployed infrastructure
- Monitor and analyze security events using centralized logging and detection solutions
- Validate the effectiveness of the implemented cybersecurity controls through attack simulations and comparative security assessments

Architecture Overview



Technologies Used:

- GNS3 (network emulation)
- pfSense (firewall & routing)
- Ubuntu Server (Linux services)
- Windows Server (enterprise services)
- Cisco/Moxa switch simulation
- Security tools: Nmap, Hydra, Fail2Ban, Suricata

Technical Stack

1. Virtualization & Network Emulation

- GNS3 (Network topology simulation)
- VirtualBox / VMware (Virtual machines hosting)
- Docker (Security tools and monitoring services)

2. Operating Systems

- Ubuntu Server (Linux services and hardening)
- Windows Server (Enterprise services and security policies)

3. Network Infrastructure

- pfSense (Firewall, routing, network segmentation, access control)
- Cisco Switch (Network switching, port security, VLAN simulation)
- Moxa Switch (Industrial OT network simulation)

4. Cybersecurity Tools

- Nmap (Network scanning and reconnaissance simulation)
- Hydra (Authentication attack simulation)
- Fail2Ban (Intrusion prevention mechanism)
- Suricata / Snort (Intrusion Detection/Prevention System)

5. Monitoring & Logging

- Wazuh or ELK Stack (Security event monitoring and log analysis)
- Grafana (Visualization of system and network activity)

Roadmap / Timeline

Phase 1: Discovery & Environment Setup

Goal: Build the foundation of the project

Tasks:

- Study OT cybersecurity concepts and hardening principles
- Install and configure GNS3 environment
- Deploy virtual machines (Windows Server, Linux Server)
- Configure pfSense firewall
- Establish basic network connectivity between components
- Design initial network architecture diagram

Deliverables:

- Functional emulated OT lab
- Network topology diagram
- Basic connectivity validation (ping, SSH, RDP)

Phase 2: System & Network Hardening

Goal: Secure the infrastructure and reduce attack surface

Tasks:

- Apply Linux hardening (SSH, firewall, services, permissions)
- Apply Windows Server hardening (GPO, RDP, accounts, services)
- Configure pfSense firewall rules and segmentation
- Implement basic switch security (port security, VLANs if applicable)
- Align configurations with CIS Benchmarks

Deliverables:

- Hardened Linux and Windows systems
- Secure network configuration (firewall + segmentation)
- Hardening documentation

Phase 3: Attack Simulation & Validation

Goal: Validate security through real attack scenarios

Tasks:

- Simulate network attacks (nmap scanning, brute force, unauthorized access)
- Test system resilience before and after hardening
- Identify vulnerabilities and verify mitigation effectiveness
- Record attack scenarios and results

Deliverables:

- Attack simulation reports
- Before/after security comparison
- Vulnerability analysis document

Phase 4: Monitoring & Security Enhancement

Goal: Improve visibility and detection capabilities

Tasks:

- Deploy monitoring stack (Docker-based environment)
- Configure IDS/IPS (Suricata or Snort)
- Centralize logs (Wazuh / ELK Stack if used)
- Create dashboards for security monitoring
- Analyze detected events and alerts

Deliverables:

- Monitoring dashboard
- IDS/IPS configuration
- Security event logs and analysis

Phase 5: Finalization & Documentation

Goal: Final validation and reporting

Tasks:

- Optimize and clean architecture
- Define Linux patch management strategy (including rollback)
- Prepare migration plans (Windows/Linux systems)
- Finalize technical documentation
- Prepare presentation and demo scenario

Deliverables:

- Final internship report
- Patch management strategy
- Migration plans
- Final presentation slides
- Demonstration of full attack → defense cycle

Expected Outcomes

At the end of this internship, the following outcomes are expected:

- A fully functional emulated OT cybersecurity environment built using GNS3, representing a realistic industrial network architecture.
- A hardened infrastructure with reduced attack surface, achieved through system and network security configurations applied on Linux, Windows Server, pfSense, and network devices.
- A segmented and secured network architecture implementing defense-in-depth principles and access control policies.
- A validated security model demonstrating the effectiveness of hardening techniques against real-world cyberattack scenarios.
- Successful simulation of cyberattacks (reconnaissance, brute force, unauthorized access) with documented results before and after mitigation.
- A monitoring and detection layer providing visibility into network activities and security events.
- A documented cybersecurity methodology including hardening procedures, attack simulations, and mitigation strategies.
- A complete technical report and final presentation summarizing the architecture, implementation, testing, and results of the project.

Methodology

The project follows a structured **cybersecurity** engineering approach based on iterative improvement and validation. **The methodology** is designed to ensure that each security mechanism is **implemented, tested, and evaluated** in a controlled environment.

Iterative Security Lifecycle

The project is based on the following cycle:

Design → Deploy → Harden → Attack → Detect → Validate → Improve

This cycle is repeated throughout the different phases of the internship to progressively strengthen the security posture of the emulated OT environment.

Phase-Based Approach

The work is divided into clearly defined phases:

- Design Phase: Definition of the **OT** architecture, network topology, and security strategy
- Deployment Phase: Setup of virtual machines, network components, and tools using **GNS3** and virtualization platforms
- Hardening Phase: Application of security configurations on operating systems and network devices
- Attack Simulation Phase: Execution of controlled **cyberattack** scenarios to test system **resilience**
- Detection & Monitoring Phase: Analysis of **logs** and alerts using security **monitoring tools**
- Validation Phase: Comparison of system behavior **before** and **after** hardening to measure effectiveness

Security Principles Applied

The methodology is guided by key cybersecurity principles:

- Defense in Depth: Multiple layers of security controls across network and system levels
- Least Privilege: Restricting access rights to the minimum required
- Attack Surface Reduction: Disabling unnecessary services and reducing exposure
- Segmentation: Separating network zones to limit lateral movement
- Continuous Validation: Testing security mechanisms through simulated attacks

Validation Strategy

The effectiveness of the implemented security measures is evaluated through:

- Comparison of system behavior before and after hardening
- Analysis of attack success/failure scenarios
- Monitoring of security logs and alerts
- Verification of access control and network restrictions